Money Plot Labs Technical Whitepaper
The Pure Math of Financial Freedom

Money Plot Labs

June 2026

Abstract

This whitepaper derives the closed-form analytical solution for a fixed-horizon retirement timeline. We present a simple linear model first to establish a transparent baseline using basic algebra. We then expand this model into exponential growth curves of an accumulation and decumulation phase. This isolates the exact number of required working years (w) as a function of savings rate, initial capital, investment growth, and desired legacy floors.

1 Introduction & Horizon Framework

Standard personal finance paradigms frequently rely on generalized heuristics and rules of thumb. While these baseline approximations offer valuable practical guidance, they often lack the multi-variable flexibility required to model specific, customized boundary conditions. To analyze these variables systematically, we establish a deterministic lifecycle framework. We define the total planning horizon (H) spanning from an initial evaluation age (T_{start}) to a maximum life expectancy threshold (T_{end}) as:

$$H = T_{\text{end}} - T_{\text{start}} \tag{1}$$

Within this horizon H, an individual works for w consecutive years and spends the remaining f years in full retirement, such that:

$$H = w + f \tag{2}$$

1.1 Parameter Definitions

- I: Annual Earned Income ($I = s + c$) ($)
- s: Annual Savings Flow ($)
- c: Annual Consumption Outflow ($)
- r: Inflation-Adjusted Real Growth Rate
- A_0: Initial Net Worth Starting Balance ($)
- w: Accumulation Horizon (Working Years)
- f: Decumulation Horizon (Freedom Years)
- L: Desired Legacy Floor Target ($)

1.2 A Note on Purchasing Power and Monetary Units

To preserve absolute analytical clarity across a multi-decade planning horizon, all currency vectors within this framework are evaluated strictly in **Real Terms (Today's Dollars)**. By adjusting our cash flow velocities (s, c) and investment yields (r) for inflation from Day 1, we isolate changes in true purchasing power. This completely removes nominal monetary noise from the geometry, ensuring that a flat retirement consumption line reflects a perfectly stable, un-degraded standard of living from the initial starting age to the terminal horizon milestone.

2 The Linear Baseline Case ($r = 0\%$)

Before introducing the compounding curves of financial markets, we establish intuitive baselines by assuming real asset growth perfectly tracks inflation ($r = 0\%$). In this environment, wealth accumulates and depletes along perfectly straight lines, making it easy to isolate how system parameters structurally shift timeline horizons.

2.1 Case 1: The Idealized Zero-Baseline Framework ($A_0 = 0, L = 0$)

We begin with a simplified world containing a zero starting balance ($A_0 = 0$) and no terminal inheritance target ($L = 0$).

During the working phase, assets grow linearly via cash savings flow s injected at the end of each period for w years:

$$\text{Peak Assets Case 1} = s \cdot w \tag{3}$$

Upon entering retirement, accumulation stops decumulation begins draining our peak nest egg by a flat annual consumption c for the remaining f retirement years:

$$\text{Peak Assets Required Case 1} = c \cdot f \tag{4}$$

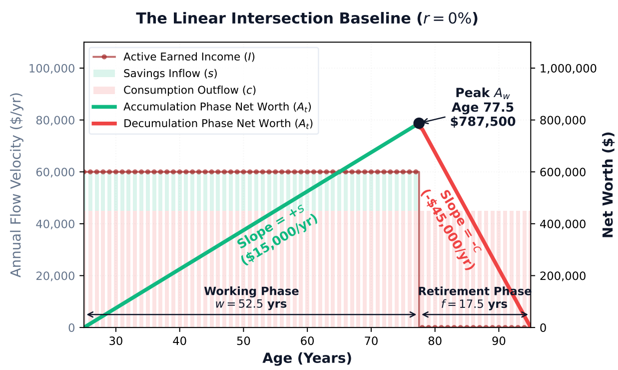

Figure 1: The straight-line boundary crossover mapping where $r = 0\%$. Assets accumulate smoothly via savings and deplete symmetrically via consumption until hitting the legacy boundary.

To locate the precise intersection boundary where working ends and retirement begins, we enforce our system constraint where the peak accumulated capital must exactly equal the total retirement drawdown:

$$s \cdot w = c \cdot f \tag{5}$$

Substituting our timeline identity $f = H - w$ allows us to solve for w:

$$s \cdot w = c \cdot (H - w) \tag{6}$$

Grouping terms containing our target variable w onto the left-hand side and factoring yields:

$$w(s + c) = c \cdot H \tag{7}$$

Dividing through isolates our fundamental idealized baseline relationship:

$$w = \frac{c \cdot H}{s + c} = \frac{c \cdot H}{I} \tag{8}$$

In the example in Figure 1, a person saving 25% of their income: $I = \$60,000$/year, $c = \$45,000$/year, $s = \$15,000$/year, $T_{start} = 25$ years, $T_{end} = 95$ years:

$$w = \frac{\$45,000 \cdot 70 \text{ years}}{\$60,000}$$
$$w = 52.5 \text{ years}$$

Another way to look at it is that the total area of the consumption bar chart should equal the area under the earned income line.

$$I * w = c * H \tag{9}$$

Equivalently, the area of the savings bars is equal to the post retirement consumption bars.

2.2 Case 2: The Parameterized Linear Solution with Boundary Modifiers ($A_0 \neq 0, L \neq 0$)

We now expand our straight-line environment to account for real-world initial endowments and estate conservation goals. We reintroduce the asset head-start parameter (A_0) and the designated capital legacy floor (L).

Incorporating these conditions alters our primary boundary accumulation and decumulation states:

$$\text{Peak Assets Case 2} = A_0 + s \cdot w \tag{10}$$

$$\text{Peak Assets Required Case 2} = L + c \cdot f \tag{11}$$

Setting our parameterized accumulation trajectory equal to the absolute terminal requirement forces the system boundary equation:

$$A_0 + s \cdot w = L + c \cdot f \tag{12}$$

Substituting our timeline identity $f = H - w$ and isolating our targeted timeline variables onto the left-hand side of the identity yields:

$$s \cdot w + c \cdot w = c \cdot H + L - A_0 \tag{13}$$

Factoring out w collects our parameters into clean operational dimensions:

$$w(s + c) = c \cdot H + L - A_0 \tag{14}$$

Dividing through by our total structural velocity $(s + c)$ delivers the complete **Parameterized Linear Equation**:

$$w = \frac{c \cdot H + L - A_0}{s + c} \tag{15}$$

This intermediate step clearly demonstrates the pure algebraic translation of our boundary mechanics: an initial balance (A_0) directly subtracts from required working years, while a desired legacy floor (L) extends the required labor timeline.

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3 Phase 1: The Exponential Accumulation Engine

When we introduce real market investment returns ($r > 0\%$), our straight lines transform into curves. During the working phase (w), net worth grows via compound interest on the initial principal while absorbing an ordinary annuity flow of annual savings c injected at the end of each period. The net worth at the exact boundary of retirement (A_w) is defined by:

$$A_w = A_0(1 + r)^w + s \cdot \frac{(1 + r)^w - 1}{r} \tag{16}$$

3.1 Step-by-Step Inductive Derivation of A_w

To understand the structural origin of the accumulation equation, we trace the boundary state changes sequentially at the close of each annual compounding cycle:

$$\begin{aligned}
A_1 &= A_0(1+r) + s \\
A_2 &= A_1(1+r) + s = [A_0(1+r) + s](1+r) + s \\
&= A_0(1+r)^2 + s(1+r) + s \\
A_3 &= A_2(1+r) + s = \left[A_0(1+r)^2 + s(1+r) + s\right](1+r) + s \\
&= A_0(1+r)^3 + s(1+r)^2 + s(1+r) + s
\end{aligned}$$

Extending this inductive sequence to a generalized working timeline of w years yields:

$$A_w = A_0(1+r)^w + s\sum_{k=0}^{w-1}(1+r)^k \tag{17}$$

The summation block represents a classic geometric progression. Applying the standard Geometric Series Summation Theorem where the common ratio is $(1+r)$, the series condenses analytically:

$$\sum_{k=0}^{w-1}(1+r)^k = \frac{(1+r)^w - 1}{(1+r) - 1} = \frac{(1+r)^w - 1}{r} \tag{18}$$

Substituting Equation 18 back into Equation 17 yields our verified final accumulation master expression matching Equation 16.

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4 Phase 2: The Exponential decumulation Engine

Upon entering retirement, the cash flow behavior flips. To match real-world physical constraints, living expenses must be accessible at the *beginning* of each calendar year. Therefore, retirement spending is modeled strictly as an **Annuity Due**.

$$R_f = A_w(1+r)^f - c \cdot \frac{(1+r)^{f+1} - (1+r)}{r} \tag{19}$$

4.1 Step-by-Step Inductive Derivation of the decumulation Phase

We trace the remaining capital balance R at the close of each retirement year sequentially, demonstrating why the timing mismatch injects an extra factor of $(1+r)$ compared to an ordinary annuity:

$$\begin{aligned}
R_1 &= (A_w - c)(1+r) = A_w(1+r) - c(1+r) \\
R_2 &= (R_1 - c)(1+r) = [A_w(1+r) - c(1+r) - c](1+r) \\
&= A_w(1+r)^2 - c(1+r)^2 - c(1+r) \\
R_3 &= (R_2 - c)(1+r) = \left[A_w(1+r)^2 - c(1+r)^2 - c(1+r) - c\right](1+r) \\
&= A_w(1+r)^3 - c(1+r)^3 - c(1+r)^2 - c(1+r)
\end{aligned}$$

Extending this progression to a total retirement duration of f years yields:

$$R_f = A_w(1+r)^f - c\sum_{k=1}^{f}(1+r)^k \tag{20}$$

Factoring out a common term of $(1+r)$ isolates a standard geometric progression inside the summation block:

$$\sum_{k=1}^{f}(1+r)^k = (1+r)\sum_{k=0}^{f-1}(1+r)^k = (1+r)\cdot\frac{(1+r)^f-1}{r} \tag{21}$$

Distributing the factored $(1+r)$ back across the numerator yields the completed, time-synchronized decumulation engine expression matching Equation 19.

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5 The Unified Derivation with Legacy Capital Floors

To complete our global solution, we enforce our final boundary constraint where our asset base at the end of life must exactly match our legacy target ($R_f = L$). By substituting our timeline identity $f = H - w$, we can map our accumulation peak directly into our decumulation model.

5.1 The Color-Coded Substitution

We inject the accumulation boundary definition (Equation 16) directly into our synchronized decumulation phase expression (Equation 19), creating a unified timeline equation:

$$\left[A_0(1+r)^w + s\cdot\frac{(1+r)^w-1}{r}\right](1+r)^{H-w} - c\cdot\frac{(1+r)^{H-w+1}-(1+r)}{r} = L \tag{22}$$

5.2 The Algebraic Isolation

Distributing the central exponential term $(1+r)^{H-w}$ across our accumulation bracket allows the localized w variables to elegantly combine and drop out of the leading terms:

$$A_0(1+r)^H + \frac{s}{r}(1+r)^H - \frac{s}{r}(1+r)^{H-w} - c\cdot\frac{(1+r)^{H-w+1}-(1+r)}{r} = L \tag{23}$$

Multiplying the entire equation by our common denominator r removes fractional complexity:

$$rA_0(1+r)^H + s(1+r)^H - s(1+r)^{H-w} - c(1+r)^{H-w+1} + c(1+r) = rL \tag{24}$$

Grouping the terms containing our targeted timeline variable w onto the right-hand side, and moving our legacy parameter to the left-hand side:

$$(1+r)^H(rA_0+s) + c(1+r) - rL = [s + c(1+r)](1+r)^{H-w} \tag{25}$$

Isolating our exponential timeline block completely:

$$(1+r)^{H-w} = \frac{(rA_0+s)(1+r)^H + c(1+r) - rL}{s+c(1+r)} \tag{26}$$

Taking the natural logarithm (ln) of both sides frees our timeline variable from the exponent:

$$(H - w)\ln(1 + r) = \ln\left[\frac{(rA_0 + s)(1 + r)^H + c(1 + r) - rL}{s + c(1 + r)}\right] \tag{27}$$

Isolating w yields the final closed-form analytical solution for the required working timeline:

$$w = H - \frac{\ln\left[\frac{(rA_0+s)(1+r)^H+c(1+r)-rL}{s+c(1+r)}\right]}{\ln(1 + r)} \tag{28}$$

Observe that evaluating the limit of Equation 28 as $r \to 0$ collapses perfectly into our baseline parameterized linear case derived in Equation 15, confirming absolute mathematical symmetry across all domains.

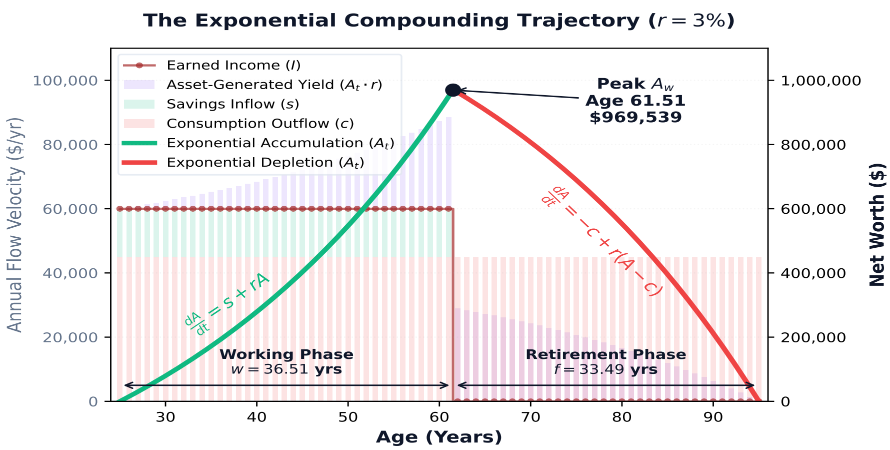

Figure 2: The exponential compounding lifecycle curves where $r > 0\%$. The accumulation phase curves upward via geometric expansion and transitions to an accelerated annuity-due decumulation phase. The total area of the consumption bar chart should equal the area under the earned income line + the total area under the earned income. The area of the savings bar chart is equal to the post retirement consumption bar chart.

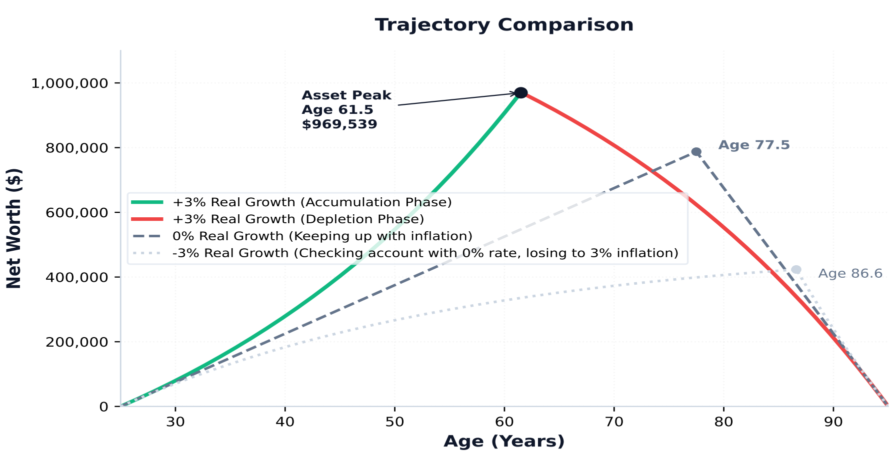

Figure 3: The exponential compounding curve compared to both assets just keeping with inflation and assets losing to inflation.